



Red Hat OpenShift AI Self-Managed 2.25

Working on data science projects

Organize your work in projects and workbenches, create and collaborate on notebooks, train and deploy models, configure model servers, and implement pipelines

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Abstract

Organize your work in projects and workbenches, create and collaborate on notebooks, train and deploy models, configure model servers, and implement pipelines.

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PREFACE

As a data scientist, you can organize your data science work into a single project. A data science project in OpenShift AI can consist of the following components:

Workbenches

Creating a workbench allows you to work with models in your preferred IDE, such as JupyterLab.

Cluster storage

For data science projects that require data retention, you can add cluster storage to the project.

Connections

Adding a connection to your project allows you to connect data inputs to your workbenches.

Pipelines

Standardize and automate machine learning workflows to enable you to further enhance and deploy your data science models.

Models and model servers

Deploy a trained data science model to serve intelligent applications. Your model is deployed with an endpoint that allows applications to send requests to the model.

CHAPTER 1. USING DATA SCIENCE PROJECTS

1.1. CREATING A DATA SCIENCE PROJECT

To implement a data science workflow, you must create a project. In OpenShift, a project is a Kubernetes namespace with additional annotations, and is the main way that you can manage user access to resources. A project organizes your data science work in one place and also allows you to collaborate with other developers and data scientists in your organization.

Within a project, you can add the following functionality:

- Connections so that you can access data without having to hardcode information like endpoints or credentials.
- Workbenches for working with and processing data, and for developing models.
- Deployed models so that you can test them and then integrate them into intelligent applications. Deploying a model makes it available as a service that you can access by using an API.
- Pipelines for automating your ML workflow.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have the appropriate roles and permissions to create projects.

Procedure

1. From the OpenShift AI dashboard, select **Data science projects**.
The **Data science projects** page shows a list of projects that you can access. For each user-requested project in the list, the **Name** column shows the project display name, the user who requested the project, and the project description.
2. Click **Create project**.
3. In the **Create project** dialog, update the **Name** field to enter a unique display name for your project.
4. Optional: If you want to change the default resource name for your project, click **Edit resource name**.
The resource name is what your resource is labeled in OpenShift. Valid characters include lowercase letters, numbers, and hyphens (-). The resource name cannot exceed 30 characters, and it must start with a letter and end with a letter or number.

Note: You cannot change the resource name after the project is created. You can edit only the display name and the description.
5. Optional: In the **Description** field, provide a project description.
6. Click **Create**.

Verification

- A project details page opens. From this page, you can add connections, create workbenches, configure pipelines, and deploy models.

1.2. UPDATING A DATA SCIENCE PROJECT

You can update the project details by changing the project name and description.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a data science project.

Procedure

1. From the OpenShift AI dashboard, click **Data science projects**.
The **Data science projects** page opens.
2. Click the action menu (⋮) beside the project whose details you want to update and click **Edit project**.
The **Edit project** dialog opens.
3. Optional: Edit the **Name** field to change the display name for your project.
4. Optional: Edit the **Description** field to change the description of your project.
5. Click **Update**.

Verification

- You can see the updated project details on the **Data science projects** page.

1.3. DELETING A DATA SCIENCE PROJECT

You can delete data science projects so that they do not appear on the OpenShift AI **Data science projects** page when you no longer want to use them.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a data science project.

Procedure

1. From the OpenShift AI dashboard, click **Data science projects**.
The **Data science projects** page opens.
2. Click the action menu (⋮) beside the project that you want to delete and then click **Delete project**.
The **Delete project** dialog opens.
3. Enter the project name in the text field to confirm that you intend to delete it.

4. Click **Delete project**.

Verification

- The data science project that you deleted is no longer displayed on the **Data science projects** page.
- Deleting a data science project deletes any associated workbenches, data science pipelines, cluster storage, and connections. This data is permanently deleted and is not recoverable.

CHAPTER 2. USING PROJECT WORKBENCHES

2.1. CREATING A WORKBENCH AND SELECTING AN IDE

A workbench is an isolated area where you can examine and work with ML models. You can also work with data and run programs, for example to prepare and clean data. While a workbench is not required if, for example, you only want to service an existing model, one is needed for most data science workflow tasks, such as writing code to process data or training a model.

When you create a workbench, you specify an image (an IDE, packages, and other dependencies). Supported IDEs include JupyterLab, code-server, and RStudio (Technology Preview).

The IDEs are based on a server-client architecture. Each IDE provides a server that runs in a container on the OpenShift cluster, while the user interface (the client) is displayed in your web browser. For example, the Jupyter workbench runs in a container on the Red Hat OpenShift cluster. The client is the JupyterLab interface that opens in your web browser on your local computer. All of the commands that you enter in JupyterLab are executed by the workbench. Similarly, other IDEs like code-server or RStudio Server provide a server that runs in a container on the OpenShift cluster, while the user interface is displayed in your web browser. This architecture allows you to interact through your local computer in a browser environment, while all processing occurs on the cluster. The cluster provides the benefits of larger available resources and security because the data being processed never leaves the cluster.

In a workbench, you can also configure connections (to access external data for training models and to save models so that you can deploy them) and cluster storage (for persisting data). Workbenches within the same project can share models and data through object storage with the data science pipelines and model servers.

For data science projects that require data retention, you can add container storage to the workbench you are creating.

Within a project, you can create multiple workbenches. When to create a new workbench depends on considerations, such as the following:

- The workbench configuration (for example, CPU, RAM, or IDE). If you want to avoid editing the configuration of an existing workbench's configuration to accommodate a new task, you can create a new workbench instead.
- Separation of tasks or activities. For example, you might want to use one workbench for your Large Language Models (LLM) experimentation activities, another workbench dedicated to a demo, and another workbench for testing.

2.1.1. About workbench images

A workbench image is preinstalled with the tools and libraries that you need for model development. You can use the provided workbench images or an OpenShift AI administrator can create custom workbench images adapted to your needs.

To provide a consistent, stable platform for your model development, many provided workbench images contain the same version of Python. Most workbench images available on OpenShift AI are pre-built and ready for you to use immediately after OpenShift AI is installed or upgraded.

For information about Red Hat support of workbench images and packages, see [Red Hat OpenShift AI: Supported Configurations](#).

The following table lists the workbench images that are installed with Red Hat OpenShift AI by default.

If the preinstalled packages that are provided in these images are not sufficient for your use case, you have the following options:

- Install additional libraries after launching a default image. This option is good if you want to add libraries on an ad hoc basis as you develop models. However, it can be challenging to manage the dependencies of installed libraries and your changes are not saved when the workbench restarts.
- Create a custom image that includes the additional libraries or packages. For more information, see [Creating custom workbench images](#).



IMPORTANT

Workbench images denoted with **(Technology Preview)** in this table are not supported with Red Hat production service level agreements (SLAs) and might not be functionally complete. Red Hat does not recommend using Technology Preview features in production. These features provide early access to upcoming product features, enabling customers to test functionality and provide feedback during the development process. For more information about the support scope of Red Hat Technology Preview features, see [Technology Preview Features Support Scope](#).

Table 2.1. Default workbench images

Image name	Description
CUDA	If you are working with compute-intensive data science models that require GPU support, use the Compute Unified Device Architecture (CUDA) workbench image to gain access to the NVIDIA CUDA Toolkit. Using this toolkit, you can optimize your work by using GPU-accelerated libraries and optimization tools.
Standard Data Science	Use the Standard Data Science workbench image for models that do not require TensorFlow or PyTorch. This image contains commonly-used libraries to assist you in developing your machine learning models.
TensorFlow	TensorFlow is an open source platform for machine learning. With TensorFlow, you can build, train and deploy your machine learning models. TensorFlow contains advanced data visualization features, such as computational graph visualizations. It also allows you to easily monitor and track the progress of your models.
PyTorch	PyTorch is an open source machine learning library optimized for deep learning. If you are working with computer vision or natural language processing models, use the Pytorch workbench image.
Minimal Python	If you do not require advanced machine learning features, or additional resources for compute-intensive data science work, you can use the Minimal Python image to develop your models.
TrustyAI	Use the TrustyAI workbench image to leverage your data science work with model explainability, tracing, and accountability, and runtime monitoring. See the TrustyAI Explainability repository for some example Jupyter notebooks.

Image name	Description
code-server	With the code-server workbench image, you can customize your workbench environment to meet your needs using a variety of extensions to add new languages, themes, debuggers, and connect to additional services. Enhance the efficiency of your data science work with syntax highlighting, auto-indentation, and bracket matching, as well as an automatic task runner for seamless automation. For more information, see code-server in GitHub. NOTE: Elyra-based pipelines are not available with the code-server workbench image.
RStudio Server (Technology preview)	Use the RStudio Server workbench image to access the RStudio IDE, an integrated development environment for R, a programming language for statistical computing and graphics. For more information, see the RStudio Server site. To use the RStudio Server workbench image, you must first build it by creating a secret and triggering the BuildConfig, and then enable it in the OpenShift AI UI by editing the rstudio-rhel9 image stream. For more information, see Building the RStudio Server workbench images. IMPORTANT Disclaimer: Red Hat supports managing workbenches in OpenShift AI. However, Red Hat does not provide support for the RStudio software. RStudio Server is available through https://rstudio.org/ and is subject to RStudio licensing terms. Review the licensing terms before you use this sample workbench.

Image name	Description
CUDA - RStudio Server (Technology Preview)	Use the CUDA - RStudio Server workbench image to access the RStudio IDE and NVIDIA CUDA Toolkit. RStudio is an integrated development environment for R, a programming language for statistical computing and graphics. With the NVIDIA CUDA toolkit, you can optimize your work using GPU-accelerated libraries and optimization tools. For more information, see the RStudio Server site. To use the CUDA - RStudio Server workbench image, you must first build it by creating a secret and triggering the BuildConfig, and then enable it in the OpenShift AI UI by editing the cuda-rstudio-rhel9 image stream. For more information, see Building the RStudio Server workbench images.  IMPORTANT Disclaimer: Red Hat supports managing workbenches in OpenShift AI. However, Red Hat does not provide support for the RStudio software. RStudio Server is available through https://rstudio.org/ and is subject to RStudio licensing terms. Review the licensing terms before you use this sample workbench. The CUDA - RStudio Server workbench image contains NVIDIA CUDA technology. CUDA licensing information is available at https://docs.nvidia.com/cuda/. Review the licensing terms before you use this sample workbench.
ROCm	Use the ROCm workbench image to run AI and machine learning workloads on AMD GPUs in OpenShift AI. It includes ROCm libraries and tools optimized for high-performance GPU acceleration, supporting custom AI workflows and data processing tasks. Use this image integrating additional frameworks or dependencies tailored to your specific AI development needs.
ROCm-PyTorch	Use the ROCm-PyTorch workbench image to run PyTorch workloads on AMD GPUs in OpenShift AI. It includes ROCm-accelerated PyTorch libraries, enabling efficient deep learning training, inference, and experimentation. This image is designed for data scientists working with PyTorch-based workflows, offering integration with GPU scheduling.
ROCm-TensorFlow	Use the ROCm-TensorFlow workbench image to run TensorFlow workloads on AMD GPUs in OpenShift AI. It includes ROCm-accelerated TensorFlow libraries to support high-performance deep learning model training and inference. This image simplifies TensorFlow development on AMD GPUs and integrates with OpenShift AI for resource scaling and management.

2.1.2. Creating a workbench

When you create a workbench, you specify an image (an IDE, packages, and other dependencies). You can also configure connections, cluster storage, and add container storage.

Prerequisites

- You have logged in to Red Hat OpenShift AI.

- You have created a project.
- If you created a Simple Storage Service (S3) account outside of Red Hat OpenShift AI and you want to create connections to your existing S3 storage buckets, you have the following credential information for the storage buckets:
 - Endpoint URL
 - Access key
 - Secret key
 - Region
 - Bucket name

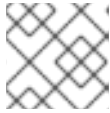
For more information, see [Working with data in an S3-compatible object store](#) .

Procedure

1. From the OpenShift AI dashboard, click **Data science projects**.
The **Data science projects** page opens.
 2. Click the name of the project that you want to add the workbench to.
A project details page opens.
 3. Click the **Workbenches** tab.
 4. Click **Create workbench**.
The **Create workbench** page opens.
 5. In the **Name** field, enter a unique name for your workbench.
 6. Optional: If you want to change the default resource name for your workbench, click **Edit resource name**.
The resource name is what your resource is labeled in OpenShift. Valid characters include lowercase letters, numbers, and hyphens (-). The resource name cannot exceed 30 characters, and it must start with a letter and end with a letter or number.
- Note:** You cannot change the resource name after the workbench is created. You can edit only the display name and the description.
7. Optional: In the **Description** field, enter a description for your workbench.
 8. In the **Workbench image** section, complete the fields to specify the workbench image to use with your workbench.
From the **Image selection** list, select a workbench image that suits your use case. A workbench image includes an IDE and Python packages (reusable code). If project-scoped images exist, the **Image selection** list includes subheadings to distinguish between global images and project-scoped images.

Optionally, click **View package information** to view a list of packages that are included in the image that you selected.

If the workbench image has multiple versions available, select the workbench image version to use from the **Version selection** list. To use the latest package versions, Red Hat recommends that you use the most recently added image.

**NOTE**

You can change the workbench image after you create the workbench.

9. In the **Deployment size** section, select one of the following options, depending on whether the hardware profiles feature is enabled.

**IMPORTANT**

The hardware profiles feature is currently available in Red Hat OpenShift AI 2.25 as a Technology Preview feature. Technology Preview features are not supported with Red Hat production service level agreements (SLAs) and might not be functionally complete. Red Hat does not recommend using them in production. These features provide early access to upcoming product features, enabling customers to test functionality and provide feedback during the development process.

For more information about the support scope of Red Hat Technology Preview features, see [Technology Preview Features Support Scope](#).

- If the hardware profiles feature is not enabled:
 - a. From the **Container size** list, select the appropriate size for the size of the model that you want to train or tune.
For example, to run the example fine-tuning job described in [Fine-tuning a model by using Kubeflow Training](#), select **Medium**.
 - b. From the **Accelerator** list, select a suitable accelerator profile for your workbench.
If project-scoped accelerator profiles exist, the **Accelerator** list includes subheadings to distinguish between global accelerator profiles and project-scoped accelerator profiles.
- If the hardware profiles feature is enabled:
 - a. From the **Hardware profile** list, select a suitable hardware profile for your workbench.
If project-scoped hardware profiles exist, the **Hardware profile** list includes subheadings to distinguish between global hardware profiles and project-scoped hardware profiles.

The hardware profile specifies the number of CPUs and the amount of memory allocated to the container, setting the guaranteed minimum (request) and maximum (limit) for both.
 - b. If you want to change the default values, click **Customize resource requests and limit** and enter new minimum (request) and maximum (limit) values.



IMPORTANT

By default, the hardware profiles feature is not enabled: hardware profiles are not shown in the dashboard navigation menu or elsewhere in the user interface. In addition, user interface components associated with the deprecated accelerator profiles functionality are still displayed. To show the **Settings → Hardware profiles** option in the dashboard navigation menu, and the user interface components associated with hardware profiles, set the **disableHardwareProfiles** value to **false** in the **OdhDashboardConfig** custom resource (CR) in OpenShift. For more information about setting dashboard configuration options, see [Customizing the dashboard](#).

10. Optional: In the **Environment variables** section, select and specify values for any environment variables.

Setting environment variables during the workbench configuration helps you save time later because you do not need to define them in the body of your workbenches, or with the IDE command line interface.

If you are using S3-compatible storage, add these recommended environment variables:

- **AWS_ACCESS_KEY_ID** specifies your Access Key ID for Amazon Web Services.
- **AWS_SECRET_ACCESS_KEY** specifies your Secret access key for the account specified in **AWS_ACCESS_KEY_ID**.

OpenShift AI stores the credentials as Kubernetes secrets in a protected namespace if you select **Secret** when you add the variable.

11. In the **Cluster storage** section, configure the storage for your workbench. Select one of the following options:
 - **Create new persistent storage** to create storage that is retained after you shut down your workbench. Complete the relevant fields to define the storage:
 - a. Enter a **name** for the cluster storage.
 - b. Enter a **description** for the cluster storage.
 - c. Select a **storage class** for the cluster storage.



NOTE

You cannot change the storage class after you add the cluster storage to the workbench.

- d. For storage classes that support multiple access modes, select an **Access mode** to define how the volume can be accessed. For more information, see [About persistent storage](#).
Only the access modes that have been enabled for the storage class by your cluster and OpenShift AI administrators are visible.
- e. Under **Persistent storage size**, enter a new size in gibibytes or mebibytes.
- **Use existing persistent storage** to reuse existing storage and select the storage from the **Persistent storage** list.

12. Optional: You can add a connection to your workbench. A connection is a resource that contains the configuration parameters needed to connect to a data source or sink, such as an object storage bucket. You can use storage buckets for storing data, models, and pipeline artifacts. You can also use a connection to specify the location of a model that you want to deploy. In the **Connections** section, use an existing connection or create a new connection:

- Use an existing connection as follows:
 - a. Click **Attach existing connections**.
 - b. From the **Connection** list, select a connection that you previously defined.
- Create a new connection as follows:
 - a. Click **Create connection**. The **Add connection** dialog opens.
 - b. From the **Connection type** drop-down list, select the type of connection. The **Connection details** section is displayed.
 - c. If you selected **S3 compatible object storage** in the preceding step, configure the connection details:
 - i. In the **Connection name** field, enter a unique name for the connection.
 - ii. Optional: In the **Description** field, enter a description for the connection.
 - iii. In the **Access key** field, enter the access key ID for the S3-compatible object storage provider.
 - iv. In the **Secret key** field, enter the secret access key for the S3-compatible object storage account that you specified.
 - v. In the **Endpoint** field, enter the endpoint of your S3-compatible object storage bucket.
 - vi. In the **Region** field, enter the default region of your S3-compatible object storage account.
 - vii. In the **Bucket** field, enter the name of your S3-compatible object storage bucket.
 - viii. Click **Create**.
 - d. If you selected **URI** in the preceding step, configure the connection details:
 - i. In the **Connection name** field, enter a unique name for the connection.
 - ii. Optional: In the **Description** field, enter a description for the connection.
 - iii. In the **URI** field, enter the Uniform Resource Identifier (URI).
 - iv. Click **Create**.

13. Click **Create workbench**.

Verification

- The workbench that you created is visible on the **Workbenches** tab for the project.

- Any cluster storage that you associated with the workbench during the creation process is displayed on the **Cluster storage** tab for the project.
- The **Status** column on the **Workbenches** tab displays a status of **Starting** when the workbench server is starting, and **Running** when the workbench has successfully started.
- Optional: Click the open icon () to open the IDE in a new window.

2.2. STARTING A WORKBENCH

You can manually start a data science project's workbench from the **Workbenches** tab on the project details page. By default, workbenches start immediately after you create them.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a data science project that contains a workbench.

Procedure

1. From the OpenShift AI dashboard, click **Data science projects**.
The **Data science projects** page opens.
2. Click the name of the project whose workbench you want to start.
A project details page opens.
3. Click the **Workbenches** tab.
4. In the **Status** column for the workbench that you want to start, click **Start**.
The **Status** column changes from **Stopped** to **Starting** when the workbench server is starting, and then to **Running** when the workbench has successfully started. * Optional: Click the open icon () to open the IDE in a new window.

Verification

- The workbench that you started is displayed on the **Workbenches** tab for the project, with the status of **Running**.

2.3. UPDATING A PROJECT WORKBENCH

If your data science work requires you to change your workbench image, container size, or identifying information, you can update the properties of your project's workbench. If you require extra power for use with large datasets, you can assign accelerators to your workbench to optimize performance.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a data science project that has a workbench.

Procedure

1. From the OpenShift AI dashboard, click **Data science projects**.
The **Data science projects** page opens.
2. Click the name of the project whose workbench you want to update.
A project details page opens.
3. Click the **Workbenches** tab.
4. Click the action menu () beside the workbench that you want to update and then click **Edit workbench**.
The **Edit workbench** page opens.
5. Update any of the workbench properties and then click **Update workbench**.

Verification

- The workbench that you updated is displayed on the **Workbenches** tab for the project.

2.4. DELETING A WORKBENCH FROM A DATA SCIENCE PROJECT

You can delete workbenches from your data science projects to help you remove Jupyter notebooks that are no longer relevant to your work.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a data science project with a workbench.

Procedure

1. From the OpenShift AI dashboard, click **Data science projects**.
The **Data science projects** page opens.
2. Click the name of the project that you want to delete the workbench from.
A project details page opens.
3. Click the **Workbenches** tab.
4. Click the action menu () beside the workbench that you want to delete and then click **Delete workbench**.
The **Delete workbench** dialog opens.
5. Enter the name of the workbench in the text field to confirm that you intend to delete it.
6. Click **Delete workbench**.

Verification

- The workbench that you deleted is no longer displayed on the **Workbenches** tab for the project.
- The custom resource (CR) associated with the workbench's Jupyter notebook is deleted.

CHAPTER 3. USING CONNECTIONS

3.1. ADDING A CONNECTION TO YOUR DATA SCIENCE PROJECT

You can enhance your data science project by adding a connection that contains the configuration parameters needed to connect to a data source or sink.

When you want to work with a very large data sets, you can store your data in an Open Container Initiative (OCI)-compliant registry, S3-compatible object storage bucket, or a URI-based repository, so that you do not fill up your local storage. You also have the option of associating the connection with an existing workbench that does not already have a connection.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a data science project that you can add a connection to.
- You have access to S3-compatible object storage, URI-based repository, or OCI-compliant registry.
- If you intend to add the connection to an existing workbench, you have saved any data in the workbench to avoid losing work.

Procedure

1. From the OpenShift AI dashboard, click **Data science projects**.
The **Data science projects** page opens.
2. Click the name of the project that you want to add a connection to.
A project details page opens.
3. Click the **Connections** tab.
4. Click **Add connection**.
5. In the **Add connection** modal, select a **Connection type**. The **OCI-compliant registry**, **S3 compatible object storage**, and **URI** options are pre-installed connection types. Additional options might be available if your OpenShift AI administrator added them.
The **Add connection** form opens with fields specific to the connection type that you selected.
6. Enter a unique name for the connection.
A resource name is generated based on the name of the connection. A resource name is the label for the underlying resource in OpenShift.
7. Optional: Edit the default resource name. Note that you cannot change the resource name after you create the connection.
8. Optional: Provide a description of the connection.
9. Complete the form depending on the connection type that you selected. For example:
 - a. If you selected **S3 compatible object storage** as the connection type, configure the connection details:

- i. In the **Access key** field, enter the access key ID for the S3-compatible object storage provider.
- ii. In the **Secret key** field, enter the secret access key for the S3-compatible object storage account that you specified.
- iii. In the **Endpoint** field, enter the endpoint of your S3-compatible object storage bucket.



NOTE

Make sure to use the appropriate endpoint format. Improper formatting might cause connection errors or restrict access to storage resources. For more information about how to format object storage endpoints, see [Overview of object storage endpoints](#).

- iv. In the **Region** field, enter the default region of your S3-compatible object storage account.
 - v. In the **Bucket** field, enter the name of your S3-compatible object storage bucket.
 - vi. Click **Create**.
- b. If you selected **URI** in the preceding step, in the **URI** field, enter the Uniform Resource Identifier (URI).
 - c. If you selected **OCI-compliant registry** in the preceding step, in the **OCI storage location** field, enter the URI.
10. Click **Add connection**.

Verification

- The connection that you added is displayed on the **Connections** tab for the project.

3.2. UPDATING A CONNECTION

You can edit the configuration of an existing connection as described in this procedure.



NOTE

Any changes that you make to a connection are not applied to dependent resources (for example, a workbench) until those resources are restarted, redeployed, or otherwise regenerated.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a data science project, created a workbench, and you have defined a connection.

Procedure

1. From the OpenShift AI dashboard, click **Data science projects**.

The **Data science projects** page opens.

2. Click the name of the project that contains the connection that you want to change.
A project details page opens.
3. Click the **Connections** tab.
4. Click the action menu () beside the connection that you want to change and then click **Edit**.
The **Edit connection** form opens.
5. Make your changes.
6. Click **Save**.

Verification

- The updated connection is displayed on the **Connections** tab for the project.

3.3. DELETING A CONNECTION

You can delete connections that are no longer relevant to your data science project.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a data science project with a connection.

Procedure

1. From the OpenShift AI dashboard, click **Data science projects**.
The **Data science projects** page opens.
2. Click the name of the project that you want to delete the connection from.
A project details page opens.
3. Click the **Connections** tab.
4. Click the action menu () beside the connection that you want to delete and then click **Delete connection**.
The **Delete connection** dialog opens.
5. Enter the name of the connection in the text field to confirm that you intend to delete it.
6. Click **Delete connection**.

Verification

- The connection that you deleted is no longer displayed on the **Connections** page for the project.

CHAPTER 4. CONFIGURING CLUSTER STORAGE

4.1. ABOUT PERSISTENT STORAGE

OpenShift AI uses persistent storage to support workbenches, project data, and model training.

Persistent storage is provisioned through OpenShift storage classes and persistent volumes. Volume provisioning and data access are determined by access modes.

Understanding storage classes and access modes can help you choose the right storage for your use case and avoid potential risks when sharing data across multiple workbenches.

4.1.1. Storage classes in OpenShift AI

Storage classes in OpenShift AI are available from the underlying OpenShift cluster. A storage class defines how persistent volumes are provisioned, including which storage backend is used and what access modes the provisioned volumes can support. For more information, see [Dynamic provisioning](#) in the OpenShift documentation.

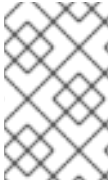
Cluster administrators create and configure storage classes in the OpenShift cluster. These storage classes provision persistent volumes that support one or more access modes, depending on the capabilities of the storage backend. OpenShift AI administrators then enable specific storage classes and access modes for use in OpenShift AI.

When adding cluster storage to your project or workbench, you can choose from any enabled storage classes and access modes.

4.1.2. Access modes

Storage classes create persistent volumes that can support different access modes, depending on the storage backend. Access modes control how a volume can be mounted and used by one or more workbenches. If a storage class allows more than one access mode, you can select the one that best fits your needs when you request storage. All persistent volumes support **ReadWriteOnce (RWO)** by default.

Access mode	Description
ReadWriteOnce (RWO) (Default)	The storage can be attached to a single workbench or pod at a time and is ideal for most individual workloads. RWO is always enabled by default and cannot be disabled by the administrator.
ReadWriteMany (RWX)	The storage can be attached to many workbenches simultaneously. RWX enables shared data access, but can introduce data risks.
ReadOnlyMany (ROX)	The storage can be attached to many workbenches as read-only. ROX is useful for sharing reference data without allowing changes.
ReadWriteOncePod (RWOP)	The storage can be attached to a single pod on a single node with read-write permissions. RWOP is similar to RWO but includes additional node-level restrictions.



NOTE

You can enable access modes that are required. A warning is displayed if you request an access mode with unknown support, but you can continue to select **Save** to create the storage class with the selected access mode.

4.1.2.1. Using shared storage (RWX)

The **ReadWriteMany (RWX)** access mode allows multiple workbenches to access and write to the same storage volume at the same time. Use **RWX** access mode for collaborative work where multiple users need to access shared datasets or project files.

However, shared storage introduces several risks:

- **Data corruption or data loss** If multiple workbenches modify the same part of a file simultaneously, the data can become corrupted or lost. Ensure your applications or workflows are designed to safely handle shared access, for example, by using file locking or database transactions.
- **Security and privacy.** If a workbench with access to shared storage is compromised, all data on that volume might be at risk. Only share sensitive data with trusted workbenches and users.

To use shared storage safely:

- Ensure that your tools or workflows are designed to work with shared storage and can manage simultaneous writes. For example, use databases or distributed data processing frameworks.
- Be cautious with changes. Deleting or editing files affects everyone who shares the volume.
- Back up your data regularly, which can help prevent data loss due to mistakes or misconfigurations.
- Limit access to RWX volumes to trusted users and secure workbenches.
- Use **ReadWriteMany (RWX)** only when collaboration on a shared volume is required. For most individual tasks, **ReadWriteOnce (RWO)** is ideal because only one workbench can write to the volume at a time.

4.2. ADDING CLUSTER STORAGE TO YOUR DATA SCIENCE PROJECT

For data science projects that require data to be retained, you can add cluster storage to the project. Additionally, you can also connect cluster storage to a specific project's workbench.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a data science project that you can add cluster storage to.

Procedure

1. From the OpenShift AI dashboard, click **Data science projects**.
The **Data science projects** page opens.
2. Click the name of the project that you want to add the cluster storage to.

A project details page opens.

3. Click the **Cluster storage** tab.
4. Click **Add cluster storage**.
The **Add cluster storage** dialog opens.
5. In the **Name** field, enter a unique name for the cluster storage.
6. Optional: In the **Description** field, enter a description for the cluster storage.
7. From the **Storage class** list, select the type of cluster storage.



NOTE

You cannot change the storage class after you add the cluster storage to the project.

8. For storage classes that support multiple access modes, select an **Access mode** to define how the volume can be accessed. For more information, see [About persistent storage](#).
Only the access modes that have been enabled for the storage class by your cluster and OpenShift AI administrators are visible.
9. In the **Persistent storage size** section, specify a new size in gibibytes or mebibytes.
10. Optional: If you want to connect the cluster storage to an existing workbench:
 - a. In the **Workbench connections** section, click **Add workbench**.
 - b. In the **Name** field, select an existing workbench from the list.
 - c. In the **Path format** field, select **Standard** if your storage directory begins with **/opt/app-root/src**, otherwise select **Custom**.
 - d. In the **Mount path** field, enter the path to a model or directory within a container where a volume is mounted and accessible. The path must consist of lowercase alphanumeric characters or -. Use / to indicate subdirectories.
11. Click **Add storage**.

Verification

- The cluster storage that you added is displayed on the **Cluster storage** tab for the project.
- A new persistent volume claim (PVC) is created with the storage size that you defined.
- The persistent volume claim (PVC) is visible as an attached storage on the **Workbenches** tab for the project.

4.3. UPDATING CLUSTER STORAGE

If your data science work requires you to change the identifying information of a project's cluster storage or the workbench that the storage is connected to, you can update your project's cluster storage to change these properties.



NOTE

You cannot directly change the **storage class** for cluster storage that is already configured for a workbench or project. To switch to a different storage class, you need to migrate your data to a new cluster storage instance that uses the required storage class. For more information, see [Changing the storage class for an existing cluster storage instance](#).

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a data science project that contains cluster storage.

Procedure

1. From the OpenShift AI dashboard, click **Data science projects**.
The **Data science projects** page opens.
2. Click the name of the project whose storage you want to update.
A project details page opens.
3. Click the **Cluster storage** tab.
4. Click the action menu (⋮) beside the storage that you want to update and then click **Edit storage**.
The **Update cluster storage** page opens.
5. Optional: Edit the **Name** field to change the display name for your storage.
6. Optional: Edit the **Description** field to change the description of your storage.
7. Optional: In the **Persistent storage size** section, specify a new size in gibibytes or mebibytes.
Note that you can only increase the storage size. Updating the storage size restarts the workbench and makes it unavailable for a period of time that is usually proportional to the size change.
8. Optional: If you want to connect the cluster storage to a different workbench:
 - a. In the **Workbench connections** section, click **Add workbench**.
 - b. In the **Name** field, select an existing workbench from the list.
 - c. In the **Path format** field, select **Standard** if your storage directory begins with **/opt/app-root/src**, otherwise select **Custom**.
 - d. In the **Mount path** field, enter the path to a model or directory within a container where a volume is mounted and accessible. The path must consist of lowercase alphanumeric characters or -. Use / to indicate subdirectories.
9. Click **Update storage**.

If you increased the storage size, the workbench restarts and is unavailable for a period of time that is usually proportional to the size change.

Verification

- The storage that you updated is displayed on the **Cluster storage** tab for the project.

4.4. CHANGING THE STORAGE CLASS FOR AN EXISTING CLUSTER STORAGE INSTANCE

When you create a workbench with cluster storage, the cluster storage is tied to a specific storage class. Later, if your data science work requires a different storage class, or if the current storage class has been deprecated, you cannot directly change the storage class on the existing cluster storage instance. Instead, you must migrate your data to a new cluster storage instance that uses the storage class that you want to use.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a workbench or data science project that contains cluster storage.

Procedure

1. Stop the workbench with the storage class that you want to change.
 - a. From the OpenShift AI dashboard, click **Data science projects**.
The **Data science projects** page opens.
 - b. Click the name of the project with the cluster storage instance that uses the storage class you want to change.
The project details page opens.
 - c. Click the **Workbenches** tab.
 - d. In the **Status** column for the relevant workbench, click **Stop**.
Wait until the **Status** column for the relevant workbench changes from **Running** to **Stopped**.
2. Add a new cluster storage instance that uses the needed storage class.
 - a. Click the **Cluster storage** tab.
 - b. Click **Add cluster storage**.
The **Add cluster storage** dialog opens.
 - c. Enter a **name** for the cluster storage.
 - d. Optional: Enter a **description** for the cluster storage.
 - e. Select the needed **storage class** for the cluster storage.
 - f. For storage classes that support multiple access modes, select an **Access mode** to define how the volume can be accessed. For more information, see [About persistent storage](#).
Only the access modes that have been enabled for the storage class by your cluster and OpenShift AI administrators are visible.
 - g. Under **Persistent storage size**, enter a size in gibibytes or mebibytes.
 - h. In the **Workbench connections** section, click **Add workbench**.

- i. In the **Name** field, select an existing workbench from the list.
 - j. In the **Path format** field, select **Standard** if your storage directory begins with **/opt/app-root/src**, otherwise select **Custom**.
 - k. In the **Mount path** field, enter the path to a model or directory within a container where a volume is mounted and accessible. For example, **backup**.
 - l. Click **Add storage**.
3. Copy the data from the existing cluster storage instance to the new cluster storage instance.
 - a. Click the **Workbenches** tab.
 - b. In the **Status** column for the relevant workbench, click **Start**.
 - c. When the workbench status is **Running**, click **Open** to open the workbench.
 - d. In JupyterLab, click **File** → **New** → **Terminal**.
 - e. Copy the data to the new storage directory. Replace `<mount_folder_name>` with the storage directory of your new cluster storage instance.


```
rsync -avO --exclude='/opt/app-root/src/___<mount_folder_name>___' /opt/app-root/src/
/opt/app-root/src/___<mount_folder_name>___/
```

For example:

```
rsync -avO --exclude='/opt/app-root/src/backup' /opt/app-root/src/ /opt/app-
root/src/backup/
```
 - f. After the data has finished copying, log out of JupyterLab.
 4. Stop the workbench.
 - a. Click the **Workbenches** tab.
 - b. In the **Status** column for the relevant workbench, click **Stop**.
Wait until the **Status** column for the relevant workbench changes from **Running** to **Stopped**.
 5. Remove the original cluster storage instance from the workbench.
 - a. Click the **Cluster storage** tab.
 - b. Click the action menu () beside the existing cluster storage instance, and then click **Edit storage**.
 - c. Under **Existing connected workbenches**, remove the workbench.
 - d. Click **Update**.
 6. Update the mount folder of the new cluster storage instance by removing it and re-adding it to the workbench.
 - a. On the **Cluster storage** tab, click the action menu () beside the new cluster storage instance, and then click **Edit storage**.

- b. Under **Existing connected workbenches**, remove the workbench.
 - c. Click **Update**.
 - d. Click the **Workbenches** tab.
 - e. Click the action menu () beside the workbench and then click **Edit workbench**.
 - f. In the **Cluster storage** section, under **Use existing persistent storage**, select the new cluster storage instance.
 - g. Click **Update workbench**.
7. Restart the workbench.
 - a. Click the **Workbenches** tab.
 - b. In the **Status** column for the relevant workbench, click **Start**.
8. Optional: The initial cluster storage that uses the previous storage class is still visible on the **Cluster storage** tab. If you no longer need this cluster storage (for example, if the storage class is deprecated), you can delete it.
9. Optional: You can delete the mount folder of your new cluster storage instance (for example, the **backup** folder).

Verification

- On the **Cluster storage** tab for the project, the new cluster storage instance is displayed with the needed storage class in the **Storage class** column and the relevant workbench in the **Connected workbenches** column.
- On the **Workbenches** tab for the project, the new cluster storage instance is displayed for the workbench in the **Cluster storage** section and has the mount path: **/opt/app-root/src**.

4.5. DELETING CLUSTER STORAGE FROM A DATA SCIENCE PROJECT

You can delete cluster storage from your data science projects to help you free up resources and delete unwanted storage space.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a data science project with cluster storage.

Procedure

1. From the OpenShift AI dashboard, click **Data science projects**.
The **Data science projects** page opens.
2. Click the name of the project that you want to delete the storage from.
A project details page opens.
3. Click the **Cluster storage** tab.

4. Click the action menu () beside the storage that you want to delete and then click **Delete storage**.
The **Delete storage** dialog opens.
5. Enter the name of the storage in the text field to confirm that you intend to delete it.
6. Click **Delete storage**.

Verification

- The storage that you deleted is no longer displayed on the **Cluster storage** tab for the project.
- The persistent volume (PV) and persistent volume claim (PVC) associated with the cluster storage are both permanently deleted. This data is not recoverable.

CHAPTER 5. MANAGING ACCESS TO DATA SCIENCE PROJECTS

5.1. GRANTING ACCESS TO A DATA SCIENCE PROJECT

To enable your organization to work collaboratively, you can grant access to your data science project to other users and groups.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a data science project.

Procedure

1. From the OpenShift AI dashboard, click **Data science projects**.
The **Data science projects** page opens.
2. From the list of data science projects, click the name of the data science project that you want to grant access to.
A project details page opens.
3. Click the **Permissions** tab.
The **Permissions** page for the project opens.
4. Provide one or more users with access to the project.
 - a. In the **Users** section, click **Add user**.
 - b. In the **Name** field, enter the user name of the user whom you want to provide access to the project.
 - c. From the **Permissions** list, select one of the following access permission levels:
 - **Admin**: Users with this access level can edit project details and manage access to the project.
 - **Contributor**: Users with this access level can view and edit project components, such as its workbenches, connections, and storage.
 - d. To confirm your entry, click **Confirm** ().
 - e. Optional: To add an additional user, click **Add user** and repeat the process.
5. Provide one or more OpenShift groups with access to the project.
 - a. In the **Groups** section, click **Add group**.
 - b. From the **Name** list, select a group to provide access to the project.

**NOTE**

If you do not have **cluster-admin** permissions, the **Name** list is not visible. Instead, an input field is displayed enabling you to configure group permissions.

- c. From the **Permissions** list, select one of the following access permission levels:
 - Admin: Groups with this access level can edit project details and manage access to the project.
 - Contributor: Groups with this access level can view and edit project components, such as its workbenches, connections, and storage.
- d. To confirm your entry, click **Confirm** ().
- e. Optional: To add an additional group, click **Add group** and repeat the process.

Verification

- Users to whom you provided access to the project can perform only the actions permitted by their access permission level.
- The **Users** and **Groups** sections on the **Permissions** tab show the respective users and groups that you granted access to.

5.2. UPDATING ACCESS TO A DATA SCIENCE PROJECT

To change the level of collaboration on your data science project, you can update the access permissions of users and groups who have access to your project.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have OpenShift AI administrator privileges or you are the project owner.
- You have created a data science project.
- You have previously shared access to your project with other users or groups.

Procedure

1. From the OpenShift AI dashboard, click **Data science projects**.
The **Data science projects** page opens.
2. Click the name of the project that you want to change the access permissions of.
A project details page opens.
3. Click the **Permissions** tab.
The **Permissions** page for the project opens.
4. Update the user access permissions to the project.

- a. Click the action menu () beside the user whose access permissions you want to update and click **Edit**.
 - b. In the **Name** field, update the user name of the user whom you want to provide access to the project.
 - c. From the **Permissions** list, update the user access permissions by selecting one of the following:
 - Admin: Users with this access level can edit project details and manage access to the project.
 - Contributor: Users with this access level can view and edit project components, such as its workbenches, connections, and storage.
 - d. To confirm the update to the entry, click **Confirm** ().
5. Update the OpenShift groups access permissions to the project.
- a. Click the action menu () beside the group whose access permissions you want to update and click **Edit**.
 - b. From the **Name** list, update the group that has access to the project by selecting another group from the list.

**NOTE**

If you do not have **cluster-admin** permissions, the **Name** list is not visible. Instead, you can configure group permissions in the input field that is displayed.

- c. From the **Permissions** list, update the group access permissions by selecting one of the following:
 - Admin: Groups with this access level can edit project details and manage access to the project.
 - Contributor: Groups with this access level can view and edit project components, such as its workbenches, connections, and storage.
- d. To confirm the update to the entry, click **Confirm** ().

Verification

- The **Users** and **Groups** sections on the **Permissions** tab show the respective users and groups whose project access permissions you changed.

5.3. REMOVING ACCESS TO A DATA SCIENCE PROJECT

If you no longer want to work collaboratively on your data science project, you can restrict access to your project by removing users and groups that you previously provided access to your project.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have OpenShift AI administrator privileges or you are the project owner.
- You have created a data science project.
- You have previously shared access to your project with other users or groups.

Procedure

1. From the OpenShift AI dashboard, click **Data science projects**.
The **Data science projects** page opens.
2. Click the name of the project that you want to change the access permissions of.
A project details page opens.
3. Click the **Permissions** tab.
The **Permissions** page for the project opens.
4. Click the action menu () beside the user or group whose access permissions you want to revoke and click **Delete**.

Verification

- Users whose access you have revoked can no longer perform the actions that were permitted by their access permission level.

CHAPTER 6. CREATING PROJECT-SCOPED RESOURCES FOR YOUR PROJECT

As an OpenShift AI user, you can access *global resources* in all OpenShift AI projects, but you can access *project-scoped resources* within the specified project only.

As a user with access permissions to a project, you can create the following types of project-scoped resources for your OpenShift AI project:

- Workbench images
- Model-serving runtimes for KServe

All resource names must be unique within a project.



NOTE

A user with access permissions to a project can create project-scoped resources for that project, as described in [Creating project-scoped resources](#).

Prerequisites

- You can access the OpenShift console.
- An OpenShift AI administrator has set the **disableProjectScoped** dashboard configuration option to **false**, as described in [Customizing the dashboard](#).
- You can access a project in the OpenShift AI console.
- You have example YAML code for the type of resource that you want to create.
You can get the YAML code from a trusted source, such as an existing project-scoped resource, a Git repository, or documentation. Alternatively, you can contact your cluster administrator to request the relevant YAML code.

Procedure

1. Log in to the OpenShift console.
2. From a trusted source, copy the YAML code that you want to use for your project resource. For example, if you can access an existing project-scoped resource in one of your projects, you can copy the YAML code as follows:
 - a. In the **Administrator** perspective, click **Home** → **Search**.
 - b. From the **Projects** list, select the appropriate project.
 - c. In the **Resources** list, search for the relevant resource type, as follows:
 - For workbench images, search for **ImageStream**.
 - For serving runtimes, search for **Template**. From the resulting list, find the templates that have the **objects.kind** specification set to **ServingRuntime**.
 - d. Select a resource, and then click the **YAML** tab.

- e. Copy the YAML content, and then click **Cancel**.
3. From the **Project** list, select your project name.
4. From the toolbar, click the + icon to open the **Import YAML** page.
5. Paste the example YAML content into the code area.
6. Edit the **metadata.namespace** value to specify the name of your project.
7. If necessary, edit the **metadata.name** value to ensure that the resource name is unique within the specified project.
8. Optional: Edit the resource name that is displayed in the OpenShift AI console, as follows:
 - For workbench images, edit the **metadata.annotations.opendatahub.io/notebook-image-name** value.
 - For serving runtimes, edit the **objects.metadata.annotations.openshift.io/display-name** value.
9. Click **Create**.

Verification

1. Log in to the OpenShift AI console.
2. Verify that the project-scoped resource is shown in the specified project:
 - For workbench images, when you create a workbench in the project, as described in [Creating a workbench](#), the workbench image that you added is available in the **Image selection** list.
 - For model-serving runtimes, see [Deploying models on the single-model serving platform](#).